1. Literature sweep, 2026-11-07: arXiv and ePrint around the `openai/math` results

Question. Which papers posted on arXiv and ePrint since 2026-09-08 bear on the program's knowledge-base entries about the ETH disproof (`KN-LIT-caa51d`) and the matrix-multiplication exponent (`KN-LIT-58d42d`, `KN-LIT-eef882`), and on ECDLP index-calculus leads?

Status of every source claim: reported by the cited paper; **not** checked here. The arithmetic in this report converts a stated 2^k to $\log_{10}$ for plotting and nothing else.

1.1. Method

- The `research-kb` MCP server and CLI were not available in this session, so the knowledge-base review used the lead atlas and repository files.
- `cryptanalysis/experiments/lit-watch` ran with a scratch configuration:
  - 60-day window, up to 50 results per query;
  - 15 arXiv queries over cs.CR, cs.CC, cs.DS, cs.SC, math.NT, math.AG and math.RA;
  - the ePrint RSS feed plus 13 ePrint searches (2 pages each).
- The fetch returned 1,601 items, with 0 failed sources.
- Ranking:
  - 225 arXiv items were dated in the window: 11 ranked as candidates, 41 as watch.
  - 1,043 ePrint search results carry no date. Their titles were scanned by hand from ID 2026/1700 onward, which roughly matches the window.
- Abstracts, and the HTML where available, were read for 16 papers. Two ePrint PDFs (2560, 2253) returned HTTP 403, so they were read from the abstract only.
- 2026/2560, 2621, 2253, 1969 and 2470 were already assessed in `research/ic_mega_pass_20261031/literature/`, and are cited from there.

1.2. Findings

1.2.1. ETH disproof (bears on KN-LIT-caa51d and LEAD-ICB-decomposition-encoding-and-sat)

- **Dias and Lima, arXiv 2611.00517.** OpenAI's general bound beats brute force 2^n only beyond $n \approx 10^{32}$ when the hidden constant is $C = 2$, 10^{126} when $C = 3$, and 10^{3517} when $C = 10$. The tensor-contraction route needs $\log_2(n+m) > 10^9$.
- **Majdoddin, arXiv 2610.41622.** Anthropic's subexponential #3-SAT algorithm crosses over at about 2^{420} clauses at density 1, lowered from 2^{33858}, and at $2^{8.6 \cdot 10^9}$ variables in general. OpenAI's Appendix D route needs about $2^{4.4 \cdot 10^{10}}$ clauses.
- **Context, not affecting any entry:**
  - MIT Hardness Group (2611.02650): ETH is false, but the Satisfiability Hypothesis remains plausible.
  - Chan et al. (2611.06813): the 3-uniform hyperclique hypothesis is false, and Max-3-SAT runs in $O(2^{0.99n})$.
  - Kozma (2610.41048): #PPM from #3SAT.
- None of these papers confirms or refutes either proof. Dias and Lima take the proofs at face value.
- **Effect.** The 2026-10-26 assessment, that the result is galactic for relation finding, now has published constants behind it. Every crossover is far above the scale of practical SAT and point-decomposition instances (Figure 1). `LEAD-ICB-decomposition-encoding-and-sat` stays `closed_obstructed`.

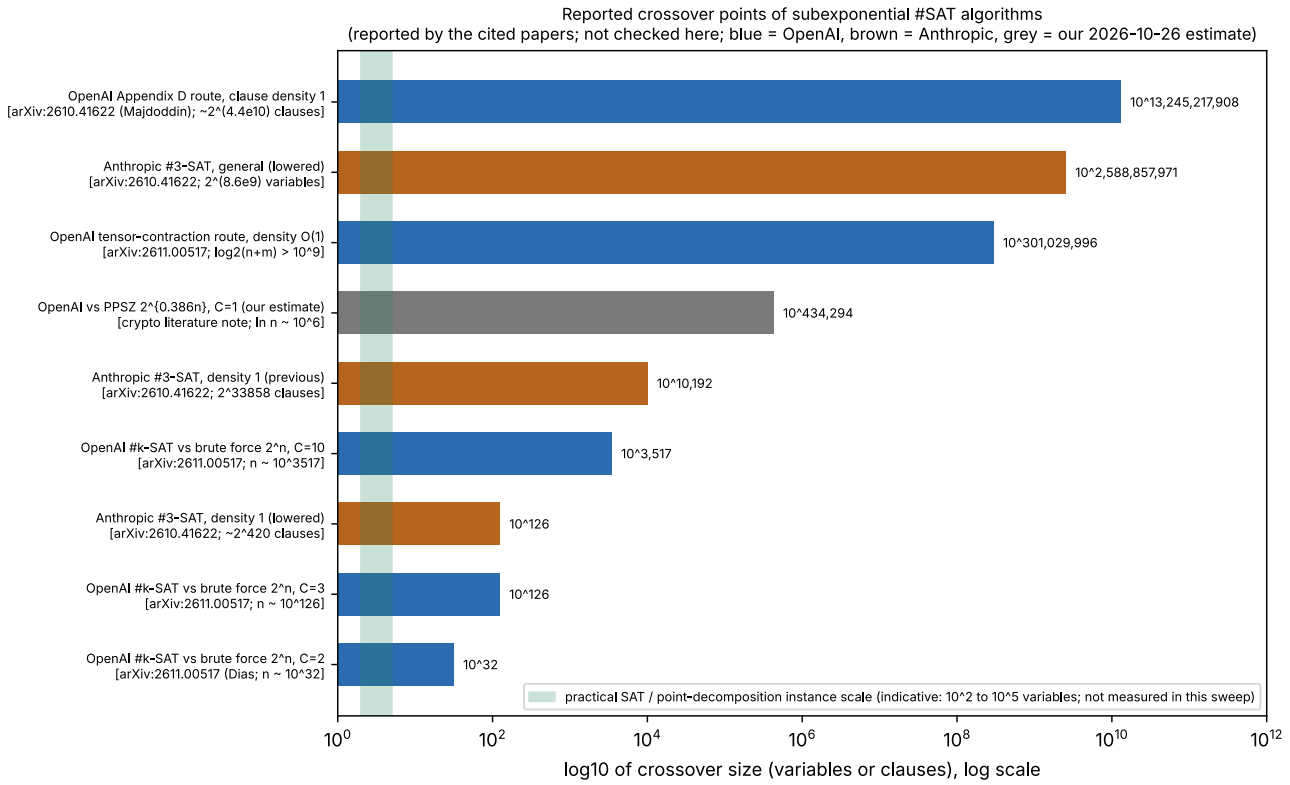


Figure 1: Crossover points as stated by the cited papers (`figures/crossovers.json`). The green band is an indicative scale of practical instances, 10^2 to 10^5 variables; it was not measured here.

1.2.2. Matrix-multiplication exponent (bears on KN-LIT-58d42d, KN-LIT-eef882 and LEAD-ICE-first-fall-degree-assumptions)

- **Lü, arXiv 2610.18914.** Corollary 4.1 gives $\omega_K \leq \frac{9}{4}$ for every algebraically closed field K: "no characteristic needs to be excluded". It also gives a cheaper shared-leg separation, with $2M - 1$ copies instead of $5M$.
  - **Derived here, standard fact:** ω depends only on the characteristic, so this covers $\mathrm{GF}(2^k)$.
  - This is a third route to $\frac{9}{4}$ in characteristic 2, alongside Amanov, van den Berg and Zuiddam (2610.30054) and Navot's Lean fork.
- **Li, arXiv 2611.04175.** An exposition of the OpenAI result; a summary reports that it argues no step needs characteristic 0.
- **Uznański, arXiv 2610.10494.** $\omega(1, k, 1) \leq 2$ for $k \leq \frac{1}{2}$, in characteristic 0 only.
- **Gao and Hu, ePrint 2026/2622.** Lattice sieving improves only if $\omega \leq \frac{9}{4} - \eta$.
- **Sun, arXiv 2610.13971** (math.GM). Claims $\omega_K \leq 2 + \frac{1}{n}$ for all $n \geq 5$ over every field, i.e. $\omega = 2$. Only the abstract was read. It is an **unverified outlier**: it changes no row and must not be used as an input.
- **Effect.** No number changes. The $\frac{9}{4}$ row stays at $2^{68.96}$ at $n = 131$, an existence-only exponent under the contested first-fall-degree model.

1.2.3. Classical ECDLP and algebraic solving

- **Xiong and Wang, ePrint 2026/2625.** Factorization over F_q in $n^{1.5532+o(1)} \log q$, against the previous $n^{1.80535}$. It is asymptotic, the abstract does not discuss root finding, and no cost row changes.
- **Budroni et al., ePrint 2026/1711** (PKP/PEP, code-based schemes): no bearing on ECDLP; discarded.
- **ePrint 2026/2469** is the program's own GPU rho paper on ECC2K-130; already recorded.
- **Already assessed on 2026-10-31** (no new action): 2026/2560 (multi-user collision walks), 2621 (polynomial approximations of dlog), 2253 (multi-instance generic bounds), 1969 (Dixon resultants), 2470 (XL for RSDP).
- **Negative result:** no new paper in the window targets ECDLP index calculus, summation polynomials or Weil descent of binary curves directly.

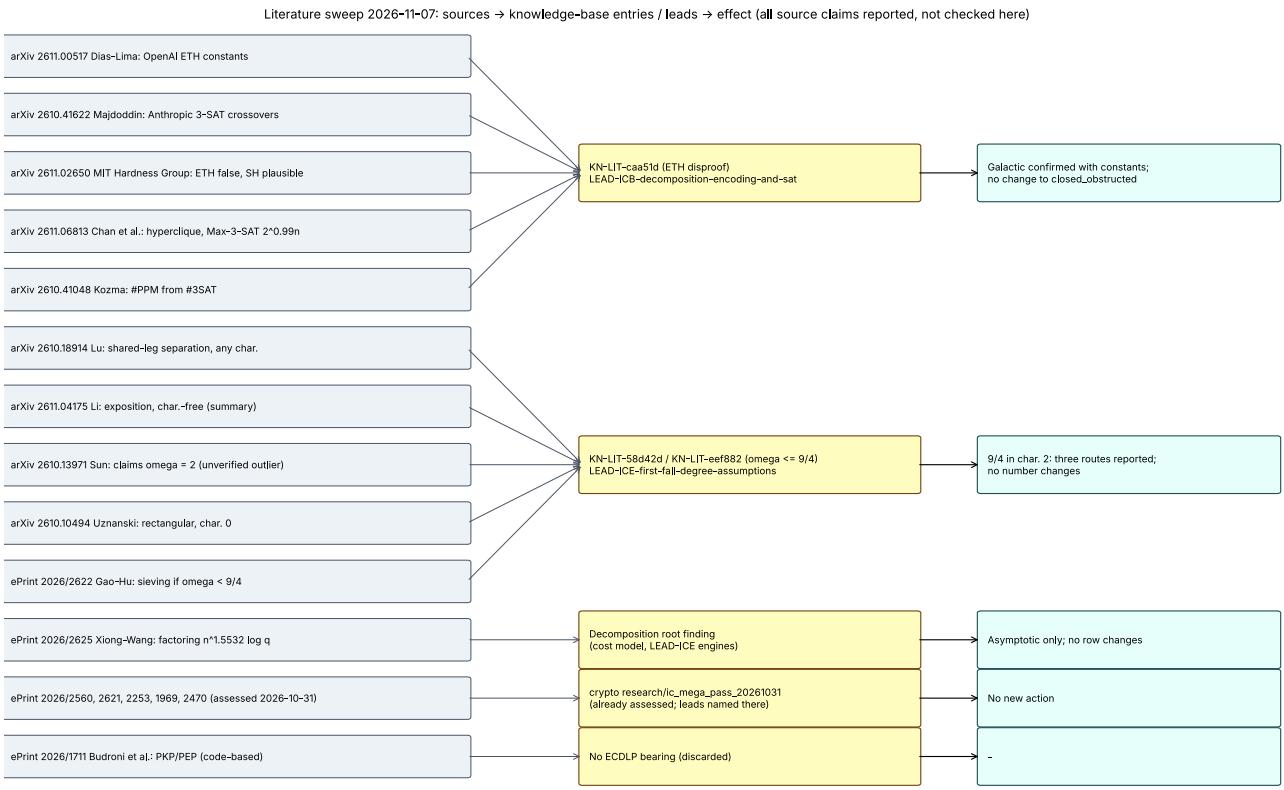


Figure 2: Sources, the knowledge-base entries and leads they bear on, and their effect.

1.3. Unresolved checks

- The OpenAI and Anthropic proofs: no third-party verdict yet.
- Build Navot's Lean fork, and check Lü's Corollary 4.1, before re-tagging `CORPUS_FAMILY_107` in `src/cryptanalysis/matmul_exponent.rs`.
- Sun's $\omega = 2$ claim: no independent review found.
- Re-run the knowledge-base queries once `research-kb` is reachable.

1.4. Sources

arXiv: 2611.00517, 2610.41622, 2611.02650, 2611.06813, 2610.41048, 2610.18914, 2611.04175, 2610.13971, 2610.10494, 2610.30054, 2610.28627. ePrint: 2026/2622, 2625, 1711, 2469, 2560, 2621, 2253, 1969, 2470. Lean: github.com/selanavot/matrix-multiplication-all-fields.